

$$\begin{array}{l}
 \left[\begin{array}{cccc} 1 & 3 & 1 & 1 \\ -4 & -9 & 2 & -1 \\ 0 & -3 & -6 & -3 \end{array} \right] \xrightarrow[\text{Row 2}]{\begin{array}{l} \text{Row 2} = \text{Row 2} \\ + 4 \times \text{Row 1} \end{array}} \left[\begin{array}{cccc} 1 & 3 & 1 & 1 \\ 0 & 3 & 6 & 3 \\ 0 & -3 & -6 & -3 \end{array} \right] \xrightarrow[\times 1/3]{\begin{array}{l} \text{Row 3} = \text{Row 3} \\ + \text{Row 2} \end{array}} \left[\begin{array}{cccc} 1 & 3 & 1 & 1 \\ 0 & 3 & 6 & 3 \\ 0 & 0 & 0 & 0 \end{array} \right] \\
 \xrightarrow[\text{Row 2}]{\text{Row 1} = \text{Row 1} -} \left[\begin{array}{cccc} 1 & 0 & -5 & -2 \\ 0 & 3 & 6 & 3 \\ 0 & 0 & 0 & 0 \end{array} \right] \xrightarrow[\times 1/3]{\text{Row 2} = \text{Row 2}} \left[\begin{array}{cccc} 1 & 0 & -5 & -2 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right]
 \end{array}$$

Verification of whether the final matrix is in rref

- Row 1 and 2 are non-zero rows. Row 3 is a zero row. $\therefore$ The condition that all non-zero rows must be above the zero rows is True.
- Row 1 Leading element = 1 in Column 1
Row 2 Leading element = 1 in Column 2
 $\therefore$ The condition that for each row, the leading element must be in a column to the right of the columns of the leading elements of all rows above it is True
- Row 2 Column 1 = Row 3 Column 1 = 0
Row 3 Column 2 = 0
 $\therefore$ The condition that for each column having a leading element, all positions below it must contain 0 is True.
- The condition that each leading element must be 1 is True.
- Row 1 Column 2 = 0
 $\therefore$ The condition that for each column having a leading element, all positions above it must contain 0 is True.
- $\therefore$ The matrix $\left[\begin{array}{cccc} 1 & 0 & -5 & -2 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right]$ is in rref.

By defn, there are no pivots in Column 3. $\therefore x_3$ is a free variable (by defn.)
The corresponding system of equations are: ① $x_1 - 5x_3 = -2$ ② $x_2 + 2x_3 = 1$
 $\therefore$ The solution set is: $x_1 = 5x_3 - 2$, $x_2 = -2x_3 + 1$, x_3 is free.
In parametric vector form, the solution set is: $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 5x_3 - 2 \\ -2x_3 + 1 \\ x_3 \end{bmatrix}$
 $= \begin{bmatrix} 5x_3 \\ -2x_3 \\ x_3 \end{bmatrix} + \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$ (Vector addition)